

WENZHI FANG

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RESEARCH BACKGROUND

Ph.D. researcher in large language models (LLMs), reinforcement learning, and distributed optimization. My work focuses on LLM post-training and reasoning, with an emphasis on RL-based alignment, agent collaboration, and communication-efficient fine-tuning for distributed on-device LLM systems.

EDUCATION

Purdue University Ph. D. Candidate in Electrical and Computer Engineering	<i>West Lafayette, IN, US</i> <i>Aug. 2023 – Present</i>
ShanghaiTech University M.S. in Communication and Information Systems	<i>Shanghai, China</i> <i>Sept. 2020 – Jul. 2023</i>
Shanghai University B.S. in Communication Engineering	<i>Shanghai, China</i> <i>Sept. 2016 – Jul. 2020</i>

WORKING EXPERIENCE

Samsung Research America (SRA), AI Center Research Intern	<i>Mountain View, CA, US</i> <i>May. 2026 – Aug. 2026</i>
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RESEARCH INTERESTS

- **Large Language Model.** Efficient Fine-Tuning of LLM & RL-based Post-training, LLM Agents
- **Machine Learning and Optimization.** Federated Learning & Distributed Optimization

SELECTED PROJECTS

ION Lab PhD Research Assistant	<i>Aug. 2023 - Present</i> <i>Advisor: Prof. Christopher G. Brinton</i> Purdue University
• Multi-Agent System Coordination, We study a multi-turn critique-and-routing controller that improves coordination among heterogeneous LLM agents by iteratively routing and refining responses while balancing agent utilization.	<i>Feb. 2026 – May. 2026</i>
• Mitigating Catastrophic Forgetting of On-device LLM via Cloud Offloading, We propose a constraint-aware RL method that embeds cloud-usage regulation into advantage computation, enabling stable, budget-controlled local–cloud LLM collaboration and reduced forgetting under continual learning without per-task reward tuning.	<i>Oct. 2025 – Feb. 2026</i>
• Collaborative Device-cloud LLMs Reasoning, We propose a unified reinforcement learning framework that empowers on-device LLMs to autonomously learn both problem-solving reasoning and routing policies, deciding when to process queries locally or to offload them to cloud LLMs. Our method removes the need for separate routing classifiers.	<i>May. 2025 – Sept. 2025</i>
• On-device LLM Fine-tuning, We propose sketching LoRA, a theoretically-grounded parameter efficient fine-tuning methodology that retains LoRA’s flexibility while adapting to the communication and computational abilities of individual devices.	<i>Aug. 2024 – Feb. 2025</i>
• Distributed Optimization for Fog Learning, We proposed an algorithm to address multi-level data heterogeneity in hierarchical federated learning, a.k.a. fog learning, deriving strong theoretical results without relying on additional data heterogeneity assumptions. This work fills a critical gap in the existing federated learning literature.	<i>Jan. 2024 – Jun. 2024</i>

Machine Learning and Optimization Lab

Aug. 2022 - Feb. 2023

Summer Intern

Advisor: Prof. [Peter Richtarik](#)

KAUST

- **Primal-Dual Optimization,**

Aug. 2022 - Feb. 2023

We studied a primal-dual hybrid gradient descent algorithm and explored variance reduced technique for primal dual optimization.

Optimization for Communication Lab

Sept. 2020 - Jun. 2022

Master Research Assistant

Advisor: Prof. [Yong Zhou](#)

ShanghaiTech

- **Zeroth-order Optimization for Federated Learning,**

Jul. 2021 - Jan. 2022

We proposed a federated zeroth-order algorithm (FedZO) with a convergence guarantee. This algorithm makes the training process forward-only, eliminating the memory overhead of backward propagation, which has been widely applied in LLM related research, e.g., LLM attack, prompt tuning, and efficient model fine-tuning.

SELECTED PUBLICATIONS

- **W. Fang**, D-J. Han, L. Yuan, and C. G. Brinton, “Bridging On-Device and Cloud LLMs for Collaborative Reasoning”, *International Conference on Machine Learning (ICML)*, 2026 [[pdf](#)]
- **W. Fang**, D-J. Han, L. Yuan, S. Hosseinalipour, and C. G. Brinton, “Federated Sketching LoRA: On-Device Collaborative Fine-Tuning of Large Language Models”, 2025, *International Conference on Machine Learning (ICML)*, 2026 [[pdf](#)]
- **W. Fang**, D-J. Han, E. Chen, S. Wang, and C. G. Brinton, “Hierarchical Federated Learning with Multi-Timescale Gradient Correction”, *Neural Information Processing Systems (NeurIPS)*, 2024. [[pdf](#)]
- **W. Fang**, D-J. Han, and C. G. Brinton, “Federated Learning over Hierarchical Wireless Networks: Training Latency Minimization via Submodel Partitioning”, *IEEE/ACM Transactions On Networking (ToN)*, 2025 [[pdf](#)]
- **W. Fang**, Z. Yu, Y. Jiang, Y. Shi, C. Jones, and Y. Zhou, “Communication-Efficient Stochastic Zeroth-Order Optimization for Federated Learning”, *IEEE Transactions on Signal Processing (TSP)*, 2022 [[pdf](#)]
- Y. Zhou, **W. Fang**, Y. Shi, and K. B. Letaief “Federated Edge Learning: Algorithms, Architectures and Trustworthiness”, *Springer Nature*, 2025 [[pdf](#)]

UNDER REVIEW

- **W. Fang**, L. Yuan, D-J. Han, G. Lan, and C. G. Brinton, “Iterative Critique-and-Routing Controller for Multi-Agent Systems with Heterogeneous LLMs”, [[pdf](#)]
- Evan Chen*, **W. Fang***, S. Wang, L. Yuan, and C. G. Brinton, “Joint Continual Learning of Local Language Models and Cloud Offloading Decisions with Budget Constraints”, 2026, [[pdf](#)]
- L. Yuan, **W. Fang**, S. Wang, and C. G. Brinton, “PAAC: Privacy-Aware Agentic Device-Cloud Collaboration”, [[pdf](#)]
- J. Lee, **W. Fang**, D-J. Han, S. Hosseinalipour, and C. G. Brinton, “Self-Play Enhancement via Advantage-Weighted Refinement in Online Federated LLM Fine-Tuning with Real-Time Feedback” [[pdf](#)]

ACADEMIC SERVICE

Reviewer of top tier conferences including NeurIPS, ICML, ICLR, AISTAT, AAAI

Reviewer of top tier journals including IEEE/ACM Transactions On Networking; IEEE Transactions on Signal Processing (TSP); IEEE Transactions On Mobile Computing (TMC); Transactions on Machine Learning Research (TMLR)

TECHNICAL SKILLS

Languages	Python, C, C++, Git, Matlab, LaTeX
Tools	PyTorch, Tensorflow, vLLM, VERL